

ANALYZING FLEXURAL STRENGTH DATA OF ICE: HOW USEFUL IS EXPLAINABLE MACHINE LEARNING?

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ABSTRACT

The climate crisis results in a rapid sea ice decline, making shipping routes accessible for longer durations throughout the year and therefore increasing maritime traffic. At the same time, ice-structure interaction is known to cause damage to ships and structures. In this context, the flexural strength is a key property of the ice. It is also an important factor in the process of the formation of ice ridges which then act as obstacles to marine transit.

Due to the complexity of natural materials, whose properties are often determined by many influencing variables, the development of a suitable material model for ice remains a major challenge. Many experimental studies on flexural strength have been done whose results can be used to draw conclusions about the material properties. However, a major drawback is that these experiments differ significantly regarding, e.g., test method, ice conditions, measured variables and testing boundary conditions, which makes a comparison as well as the derivation of general laws for material properties of ice challenging. Moreover, most studies investigate univariate relationships whereas in reality the behavior of ice is influenced by numerous factors.

In this paper an explainable machine learning approach to analyze data from various bending tests and to identify relevant features regarding the flexural strength of ice is presented. Using an approach similar to Kellner et al. [1,2], a database of flexural strength tests is established. The data is used to create machine learning models, whose predictions are interpreted with the explainable AI (XAI) method Shapley Additive exPlanations (SHAP). The goal is to show a new approach to investigate the flexural strength of ice and to get a better understanding of how suitable the use of XAI for this problem is.

Keywords: flexural strength, ice mechanics, material modeling, data analysis, machine learning, explainable AI

1. INTRODUCTION

The climate crisis results in a rapid sea ice decline which will probably continue for decades to come [3,4]. Sea areas that were previously covered by ice year-round will be (or are already) only seasonally or partially covered. Activities in these areas are increasing, particularly maritime traffic. However, ice can damage ships and structures. Hence, to protect the vulnerable Arctic wildlife [5,6], for the health and safety of operators, and due to economic reasons, safe shipping is imperative [7]. To this end, accurate ice load prediction methods and an improved understanding of ice mechanics are desirable.

In this context, ice flexural strength is a key property [8,9]. It is also important to the process of ice ridge formation. Ice ridges, in turn, can cause high loads on ships and structures, see, e.g., [10]. Various studies on the flexural strength of ice have been done, e.g., [11–17]. Most studies are univariate, i.e., they focus on the influence of one variable on flexural strength at a time, e.g., porosity [8,9] or specimen volume [18]. However, the behavior of ice can be seen as an n -dimensional space where the dimensions represent various, possibly interacting variables. The multivariate studies necessary to investigate this would require a high number of tests and are seldom done. This is reflected in prediction methods, e.g., the curve fit by [8], giving flexural strength σ_f in dependence of brine volume v_b .

$$\sigma_f = 1.76 \exp(-5.88\sqrt{v_b}) \quad (1)$$

Though Eq. (1) captures the general trend, the scatter of the underlying data (i.e., the σ_f values) is large [19], hinting at additional influencing variables or effects. A more recent study by Aly et al. [20] addressed the size effect of specimen sizes showing that the latter can have some impact on the flexural strength, but not alone as the scatter in data is still considerable.

Here, the idea is to emulate a multivariate study by using the data from various univariate studies to train a machine learning (ML) model. The ML model can possibly make more accurate predictions, e.g., through considering several input variables. In addition to accurate predictions, it is point of interest how the (black box) ML model, which has no access to physical laws, makes the predictions. An explainable artificial intelligence (XAI) method termed SHAP (SHapley Additive exPlanations) is used to explain the predictions, e.g., by ranking features by influence. Furthermore, the SHAP explanations are linked back to ice mechanical phenomena.

The explanation results and their mechanical interpretation are thought to be useful for understanding ice mechanics with respect to bending failure. In addition, the results may be used to quantify known effects, e.g., a lower porosity results in a higher flexural strength, and also support making decisions on what and how to model ice.

2. MATERIALS AND METHODS

2.1 Identification of bending and shear stress

The flexural strength of ice is commonly tested in three different setups: the cantilever beam test, the three-point-bending test, and the four-point-bending test. To derive the flexural strength values from these tests, first order beam theory is used. In addition to bending stresses, shear stresses potentially occur at the failure location and can contribute to failure. The shear stress is usually one order of magnitude lower than the flexural stress, but neither the sensitivity of ice towards shear stresses is clear [9] nor the impact of combined shear and normal stresses in the ice specimen. In four-point bending tests, pure bending occurs between the two mid-supports, whereas in three-point bending tests the shear stress has a maximum at the location of load introduction which is usually the location of failure. Therefore, the shear stress is calculated according to first order beam theory for each test and added to the analyzed data set.

2.2 Database of bending tests

A database of ice tests for investigating the flexural strength of ice was established based on studies from the literature. This includes different test setups, freshwater and saltwater ice, specimen sizes, etc., from the sources [11–17]. The raw data comprises 889 samples, of which 608 are freshwater and 281 are saltwater ice tests. The features are given in Table 1 in the Appendix. The data is tabular, with each row representing one test run and each column one feature.

In cases where brine volume v_b was not given it was calculated following [21] as

$$v_b = 0.521 S + 49.185 T_{ice} \quad (2)$$

With salinity S in parts per thousand (ppt), and T_{ice} the specimen temperature. Next, shear stresses were calculated according to [22]

$$\tau_{max} = \frac{3}{2} \frac{Q}{bh} \quad (3)$$

with b and h the beam cross-section dimensions and Q the shear force. Lastly, the beam geometry was considered through the volume and the ratio of length to thickness, the above-used b and h were not further used.

2.3 Data cleaning

The second step was the general data scrubbing, or cleaning. The foremost issue is dealing with missing values. Either the algorithm must be able to deal with missing values or suitable substitutes, e.g., average values, must be used. Here, generally the following features were excluded due to a high percentage of missing values: grain size, porosity, Young's modulus, columnar loading and the load-deflection slope. Due to the low number of data points, four-point bending tests were removed from the feature type of test. For observations where salinity was missing, the average salinity was taken as a substitute value for saltwater ice. For freshwater ice this value was simply set to zero. Since there is interest in both, predictions and their explanations, the models were trained as if applied a priori to tests. That is to say, all features whose values can only be obtained from bending test results, e.g., failure load, time until failure, loading rate and shear stress, were discarded.

2.4 Machine learning models

The features that were used are expected to exhibit numerous nonlinear relationships and the data contains missing values. This can make the application of classic statistical tools difficult, whereas machine learning (ML) models may still be useful and applicable [23]. Briefly, ML algorithms learn from data through a process termed training or learning. Training means identifying the model architecture (i.e., its parameters) that minimize the prediction error of the model for the samples in the training set.

Thereby created models can be interpreted as very complex function fits containing, e.g., numerous if-then statements. The models are then used to predict output values based on previously unseen data, i.e., new data samples. For an in-depth treatment of ML see, e.g., [24] or [25].

Here, ML models are used to predict flexural strength during bending tests of ice. As ML models, different tree-based ensemble models [26] were used for several reasons. For tabular data, tree-based algorithms often outperform other methods, at least for moderately sized datasets [27,28]. They do not assume or require the data to follow any distribution or be scaled and are robust regarding outliers [25]. Not least, preliminary studies showed tree-based models to perform well on this data and a similar dataset on ice compression tests [2].

Different tree-based algorithms were compared, based on either adaptive boosting, random forests, and gradient boosting, see also [25, 29]. As software packages scikit-learn [30], XGBoost [31] and SHAP [32, 33] were employed.

As performance metric, the mean absolute error (MAE) was mainly used.

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (4)$$

Here, y_i and $\hat{y}_i$ indicate the true and predicted flexural strength, respectively and N is the number of observations. It is straightforward to interpret as, e.g., it has the same unit as the target values (i.e., the predictions). Moreover, the distribution of flexural strength values is strongly skewed. Using a performance measure based on the sum of squared errors (as, e.g., in the root mean squared error, RMSE) would risk that possibly large errors on few observations in the tail of the distribution have a disproportional influence. [34]

A hyperparameter optimization was done by performing three grid search runs for each model. The final hyperparameters are given in Table 2 in the Appendix. Lastly, early stopping was used for the gradient-boosted tree models. That is, if the model score does not improve over a fixed number of training rounds, the training is stopped. This is a simple and efficient method against overfitting [24].

2.5 Explainability

It is not only point of interest *what* values the machine learning models predict, but also *why* the models predict specific values. There is, however, a trade-off between complexity, accuracy, and explainability of the model, see, e.g., [2, 35]. The more complex the data, the more complex the models need to be for accurate predictions. This, in turn, conflicts with knowledge discovery, i.e., explainable models. A common criticism of machine learning models is “the lack of novel laws, understanding, and knowledge arising from their use” [36]. In response to this, different models that known as explanation models have been developed, see, e.g., overview in [32, 35].

Here, two methods were used. Firstly, the *SHapley Additive eXplanation* method (SHAP) which draws a parallel between fairly attributing the payout of a game to its players and the impact of features to a model output. It is based on a solid theoretical foundation and gives consistent results [33]. At the base of this method are SHAP values which give the impact of each feature (i.e., the player) on the model output (i.e., the payout) and have the same unit as the predictions (see Appendix A2).

Secondly, a model-specific global approach is used in which the average impurity reduction (MDI) is calculated for each feature. The MDI is defined as the total decrease in node impurity, weighted by the proportion of samples reaching that node averaged over all trees of the ensemble.

2.6 Methodology

Fig. 1 gives an overview of the overall methodology. First, a tabular database that combines data from several studies from the literature is established and cleaned. The data is then used to predict the flexural strength of ice via suitable ML models (in black). These predictions are then analyzed by the explanation model (in blue) which uses both the data and the ML model as input. Thus, feature rankings, interaction effects and rare high magnitude effects can be identified. These, in turn, are thought to lead to new insights that can help to create new ice models and improve our understanding of ice mechanics.

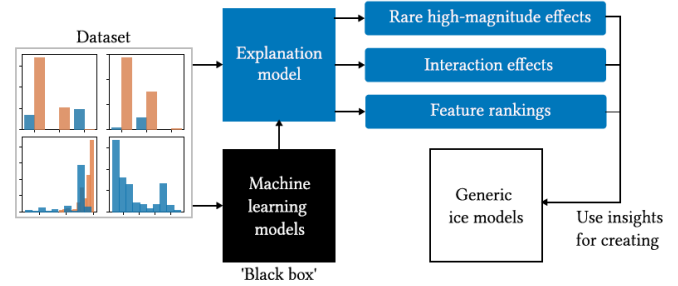


FIGURE 1: OVERVIEW OF THE METHODOLOGY. REPRODUCED FROM [2,37]

3. ANALYSIS AND RESULTS

3.1 Exploratory analysis

The exploratory analysis is the third step after obtaining and cleaning the data. Figure 2 shows a histogram of the flexural strength data. As mentioned, the data is skewed to the right. The expected value of the distribution is 1.09 MPa with a standard deviation (std) of 0.75 MPa, which is in line with the results from [19], who stated that the average flexural strength of sea ice is around 1 MPa. Since most of the used data originates from tests in freshwater, the slightly higher average value seems reasonable.

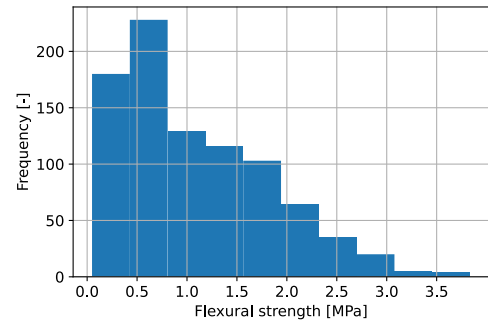


FIGURE 2: HISTOGRAM OF FLEXURAL STRENGTH DATA

3.2 Performance of ML models

Applying the explanation model only makes sense if the models perform well. To access the general prediction performance of a model, 20% of the data was restrained for testing. Furthermore, a five-fold cross validation was performed to check whether the overall model architecture suits the problem well, i.e., how good it generalizes. The results of the cross-validation showed a mean of the mean average error (MAE) of 0.323 MPa with a std of 0.036 MPa. The MAE for the testing data is 0.278 MPa, which is fairly accurate, since the relative MAE with respect to the average flexural strength is 25.5% and with respect to the total range of the flexural strength 7.3%. For this testing data, Fig. 3 shows a comparison of the true and the predicted flexural strength, i.e., the general prediction performance. The model seems to predict lower strength values with higher accuracy, while scatter increases for values above 1 MPa. There is a tendency that true strength values ranging from 0.8 MPa to 1.8 MPa are more often overestimated while true strength values above 1.8 MPa tend to be underestimated. However, more data is needed to confirm this trend.

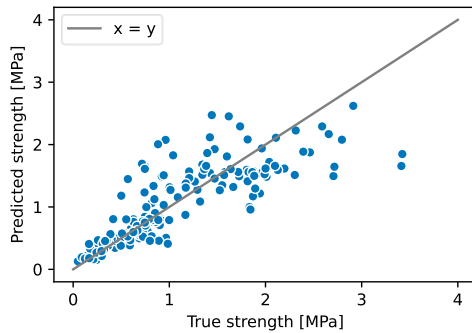


FIGURE 3: COMPARISON BETWEEN THE TRUE STRENGTH AND THE PREDICTED STRENGTH

3.3 Explanation results

Next, some selected explanation results are presented in Figs. 4 to 7. A first impression is given in Fig. 4 which gives a feature ranking for the XGBoost model based on mean SHAP values (left side). An additional feature ranking based on the MDI is given in Fig. 8 in the Appendix A.4. Comparing the results of both methods shows that such rankings can differ in magnitude and prevalence, but the overall order of feature importance is similar. In addition to the mean SHAP values for a feature, Fig. 4 shows beeswarm plots on the right side, which indicate local interpretations. In these beeswarm plots the feature value (i.e., the specific value of one feature for one observation) is color-coded, i.e., high feature values are represented in red and low feature values in blue. The horizontal axis shows the related SHAP value for one observation (i.e., one dot), whereby negative values indicate a decreasing trend in the flexural strength prediction and vice versa. Multiple observations with similar SHAP values pile up to indicate density. Grey dots indicate a

missing value, where the models tried to make a substitute decision.

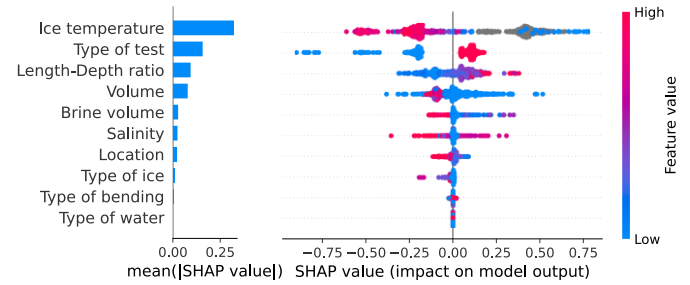


FIGURE 4: EXPLANATION RESULTS WITH SHAP

The results show that ice temperature and type of test are the most important features to predict the flexural strength of ice. According to the beeswarm plot, a higher temperature tends to lower flexural strength. The same applies to the effect of brine volume. In the beeswarm plot of the type of test, cantilever beam tests are encoded as 0 and three-point bending tests as 1, or blue and red, respectively. It can be seen that cantilever beam tests have an overall decreasing effect on the flexural strength prediction while three-point bending tests evoke the opposite effect. A high ratio between the length of the beam and the ice thickness, or short length-depth ratio, leads to higher flexural strength predictions and vice versa.

However, conclusive trends cannot be found for all features, e.g., volume and salinity. Although a high beam volume always seems to have a small decreasing effect on the flexural strength, it can also be seen that a low beam volume can result in higher and lower predictions. Similarly, high values for salinity can also cause both, higher and lower predictions, while for low salinity values, i.e., freshwater data points, salinity has almost no impact. Thus, the beeswarm plots for volume and salinity seem inconclusive yet. Fig. 5 shows the interaction effects between the ice temperature and the brine volume. In this dependence plot the Y-axis represents the impact of the ice temperature on the model output. There is a trend that lower ice temperatures have an overall increasing effect on the flexural strength prediction and vice versa. There seems to be an inflection point at an ice temperature of around -15°C . Lower ice temperatures have a positive marginal effect on the model output while higher ice temperatures lower the model output. Furthermore, vertical color patterns that could indicate interaction effect are searched for, i.e., a secondary feature amplifying or dampening the effect of the original feature. The color pattern indicates that high brine volumes can mainly be found for ice temperatures above -7°C . The blue stripes on the left side of the graph (close to the axis) represent data points with missing values for the ice temperature where the SHAP value was guessed.

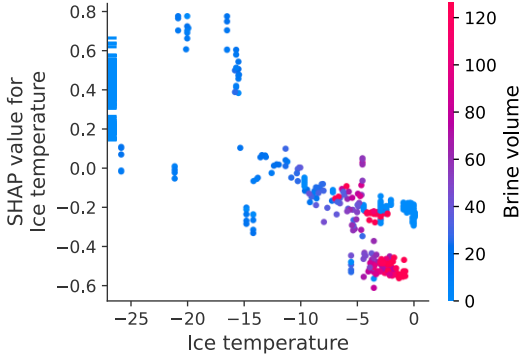


FIGURE 5: DEPENDENCE PLOT BETWEEN ICE TEMPERATURE AND BRINE VOLUME

Fig. 6 shows another dependence plot which accounts for the interaction effects between the type of bending and the type of test. Here, the downwards bending and cantilever beam test are encoded as 0 and upwards bending and three-point bending test are encoded as 1. Each bending type can lead to both, an in- and decrease of the predicted flexural strength. Furthermore, for most data points, a trend appears which suggests that the type of bending is more relevant when performing cantilever beam tests. However, this is not true for all data points (see Fig. 6, bottom right side).

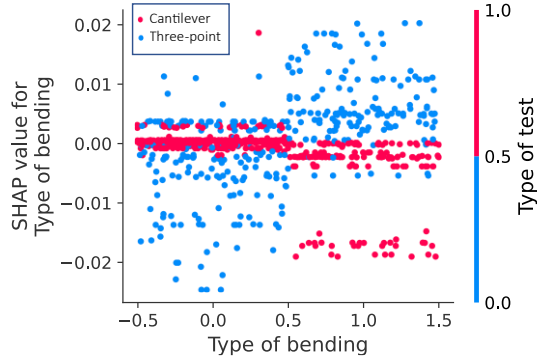


FIGURE 6: DEPENDENCE PLOT BETWEEN TYPE OF BENDING AND TYPE OF TEST

Fig. 7 shows the interaction effects between the length-depth ratio and the volume. An inflection point at a length-depth ratio of around 7 can be seen at which the length-depth ratio has least impact on the model output. For smaller length-depth ratios the predicted strength decreases while larger values cause the opposite effect. Hence, for length-depth ratios above 10, the overall impact on the model output does not appear to increase significantly. Furthermore, the impact of the length-depth ratio on the model output seems to decrease when the volume is high. However, since the length-depth ratio of most samples with a high volume is below seven, it remains to be investigated whether this is also the case for larger length-depth ratios.

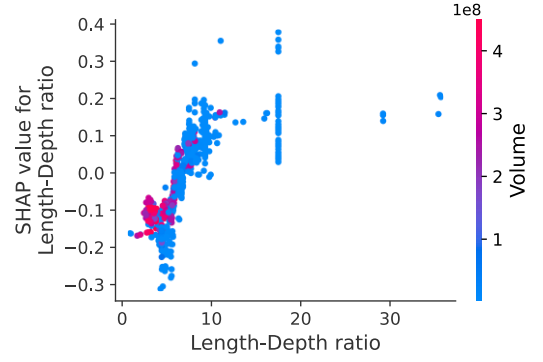


FIGURE 7: DEPENDENCE PLOT BETWEEN LENGTH-DEPTH RATIO AND VOLUME

4. DISCUSSION

The machine learning (ML) models predict flexural strength with satisfying accuracy. Additionally, the good cross-validation performance indicates that the models generalize well to previously unseen data. The choice of models appears to be well-justified.

Moreover, some features in the data base were neglected in this analysis, e.g., failure load, loading rate, shear stress etc., because these feature values can only be obtained from the conducted bending test itself and therefore show strong correlations with the target value, i.e., the flexural strength. A model that has these additional features to learn from is expected identify the strong correlation as a pattern and therefore to have an increased performance, i.e., higher prediction accuracy. In addition, more data, i.e., more samples can also increase performance and generalizability, as the results from Kellner [1,2] show, who used a similar approach to investigate the compressive strength of ice based on around 4600 observations, which is more than five times the number of observations analyzed in this study. However, this work is mainly focused on the explanation of the predictions rather than in accurate predictions.

The SHAP explanations in Fig. 4 show that high ice temperatures result in lower flexural strength predictions and vice versa. This trend is in line with general ice mechanics [38]. Fig. 5 suggests that this trend is nearly linear. Furthermore, this dependence plot indicates that the decreasing effect of high ice temperatures is amplified by large brine volumes. This result also seems reasonable because brine volume is thought to be one of the main influential factors for the flexural strength of sea ice, since large brine volume increases (together with air inclusions) the porosity, which leads to a significant strength reduction [8,9]. The results also show that most samples with large brine volume occur at rather high temperatures. This might be associated with the temperature dependency of the brine volume. At low temperatures the brine partially starts freezing, resulting in a lower brine volume and a higher brine concentration.

The results for the type of test (see Fig. 4) reveal that the flexural strength tends to be smaller when conducting cantilever tests instead of three-point bending tests. There are several

reasons that support this result. First, stress concentrations at the root of a cantilever beam can cause early failure, resulting in lower flexural strength measurements. Furthermore, size effects might come into play. Whereas the size of the ice specimen is usually limited by the laboratory set-up for three-point bending tests, there is theoretically no size limit for cantilever beam tests, apart from the stroke of the load applicator. Therefore, the beam volume is often larger for cantilever beam tests and in turn, larger ice specimens have a higher probability of flaws and cracks to be present at high-stress locations, which can lower the strength [18]. Although the beeswarm plot for volume in Fig. 4 appears to be inconclusive, a closer look reveals that only 13 samples with low volume have a decreasing effect on the model output. Most data points follow the known behavior where low volumes tend to higher flexural strengths and vice versa.

Regarding the type of bending, Fig. 6 suggests that for most data points, as already mentioned, the type of bending i.e., upwards or downwards bending, has a higher relevance for cantilever beam tests than for three-point bending. A hypothetical reason for this could be the temperature gradient, which is only present in in-situ tests, i.e., cantilever beam tests. Therefore, ice properties might differ significantly over the ice depth and the bending direction becomes more important. The results do not clearly confirm this hypothesis since there are also some data points from three-point bending tests where the type of bending seems to have a large impact, especially for upward bending. Thus, more data and research are needed to support or discard this hypothesis.

Regarding the length-depth ratio, Figs. 4 and 7 reveal that the predicted flexural strength increases when the length-depth ratio increases and vice versa. A possible explanation might be that a low length-depth ratio refers to higher shear stresses which interfere with pure bending stresses and can cause premature failure and therefore reduce the flexural strength. According to simple beam theory, the flexural strength can be expressed in dependence of shear stress, length-depth ratio and a constant c , which depends on the type of test.

$$\sigma_f = \tau_{max} \cdot \frac{l}{h} \cdot c \quad (5)$$

Yet, little is known about the effects of shear stresses on the flexural strength of ice which is why this remains subject of future research. However, in the standardized testing methods for measuring mechanical properties, Schwarz et. al [39] recommend a length-depth ratio of at least 7-10 to avoid shear stresses during cantilever beam tests. In Fig. 7 there appears to be an inflection point at a length-depth ratio of seven. Furthermore, the strong gradient suggests a high sensitivity for values below 10, i.e., small changes of the length-depth ratio cause major changes to the overall impact the feature has on the output. On the other hand, length-depth ratio values greater than 10 have an increasing effect on the flexural strength but the overall impact does not change much. Thus, the results appear to confirm the recommendation of Schwarz et al. [39], that length-depth ratios of at least 10 are desirable.

Lastly, there are various sources of error and uncertainty. In ML, the prediction error can be broken down into three parts: bias, variance, and irreducible error. The latter is introduced in the way the prediction problem is framed, e.g., it can be caused by influential but unknown variables. It cannot be controlled. Bias is, e.g., caused by erroneous assumptions. Variance originates from a high model complexity. Since tree-based models make no assumptions, they are usually not susceptible to high bias. High variance is unlikely since there are several regularization methods in place, e.g., early stopping. In all, it is not expected that the models suffer from either high bias or variance.

Regarding the explanations, these appear to make sense from an ice mechanics perspective. The results from an additional explanation method (MDI) are similar but not identical (see Appendix A.4). However, Lundberg & Lee [32] stated that the SHAP method delivers valid explanations because it is based on a strong theoretical background and the only method that fulfills all three desirable properties, i.e., local accuracy, missingness and consistency. To the authors' knowledge there is no similar study regarding the reliability of feature importance according to the MDI, which is why the SHAP explanations were mainly used.

Overall, the above-described approach appears to be valid and useful. Yet it is expected that long-term more complete data sets of measurements will be needed to substantiate the accuracy, generalizability, and explanatory power of the ML models. This would carry over to more conclusive links between ML and XAI results and ice mechanics knowledge.

5. CONCLUSION

From the analysis of flexural strength data of ice, the following conclusions can be drawn:

- For accurate predictions of flexural strength of ice, the use of ML models is a useful approach which allows to consider interaction effects between multiple influencing parameters. For quick estimates, classic empirical formulas that only consider one or two important parameters are sufficient.
- Explanation models are a good alternative for generating further insight into ice mechanics, making decisions on what and how to model ice, and quantify relationships over a large range of experimental data.
- The presented approach appears to be one way to emulate multivariate studies which are not feasible experimentally.
- It is implied that the flexural strength is affected by a number of parameters which should be measured in future experiments. This might also explain the large scatter in flexural strength values when their dependence on a single parameter is assessed.
- Nonetheless, it is regarded as a complementary approach to "classic" research methods focusing on experimental work, hypotheses and derivation of laws.

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APPENDIX

A.1 Dataset features

TABLE 1: FEATURES AND UNITS OF RAW DATA

Feature	Unit
Direction of bending	Category
Type of ice	Category
Location	Category
Columnar loading	Category
Type of test	Category
Type of water	Category
Width of specimen	mm
Thickness of specimen	mm
Length of specimen	mm
Loading rate	MPa/s
Flexural strength	MPa
Time to failure	s
Ice temperature	°C
Young's modulus	GPa
Grain size	mm
Maximum load	N
Porosity	%
Salinity	%
Brine volume	%
Load-deflection slope	-
Temperature	°C

A.2 Calculation of SHAP values

To compute SHAP values, the following holds: f is the model, $f(x)$ a prediction for input vector x . $f_x(S) = E[f(x)|x_S]$ is the expected value of f conditioned on a subset S of the input features. Furthermore, the number of input features is M and N the set of all input features. Then

$$\phi_i = \sum_{S \subseteq N/\{i\}} \frac{|S|!(M-|S|-1)!}{M!} (f_x(S \cup \{i\}) - f_x(S)) \quad (6)$$

Now, ϕ_i is the difference between $E[x_1]$ and $E[f(x)]$. In general

$$\sum_{i=0}^M \phi_i = f(x) \quad (7)$$

A.3 Hyperparameter-Tuning

TABLE 2: HYPERPARAMETER SETTINGS OF THE XGBOOST MODEL

Hyperparameter	Setting
Alpha	0
Eta	0.07
Gamma	0
Lambda	0
Max_depth	6
Min_child_weight	0
subsample	0.5

A.4 Feature Relevance according to the MDI

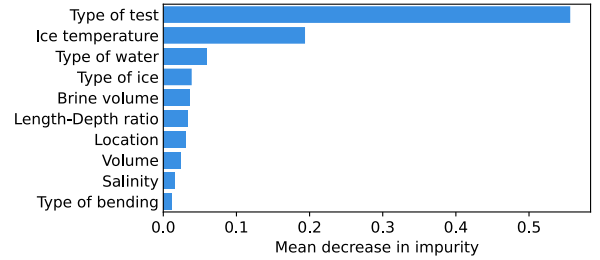


FIGURE 8: FEATURE IMPORTANCE BASED ON THE MDI

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